



PREDICTING YELP BUSINESS CLOSURES AND POST-2020 REVIEW STAR RATINGS

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WHY STUDY BUSINESS CLOSURES AND REVIEWS?

- Business closures impact both business owners as well as other community members
- It is in the interest of the business owner to keep their business afloat, and to play for potential initiatives to increase customer interaction with the business
- Rise in AI tools and large, publicly-available datasets make the process much simpler and more accessible



DATA

The paper explored the Yelp Open Dataset

- Open source
- Popular in related literature
- Only focus on the 5 JSON Files

The Dataset



6,990,280

Reviews



150,346

Businesses



11

Metropolitan areas



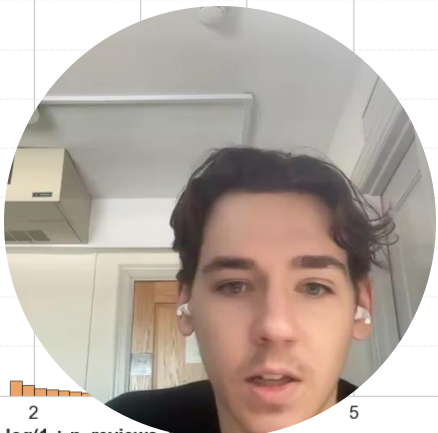
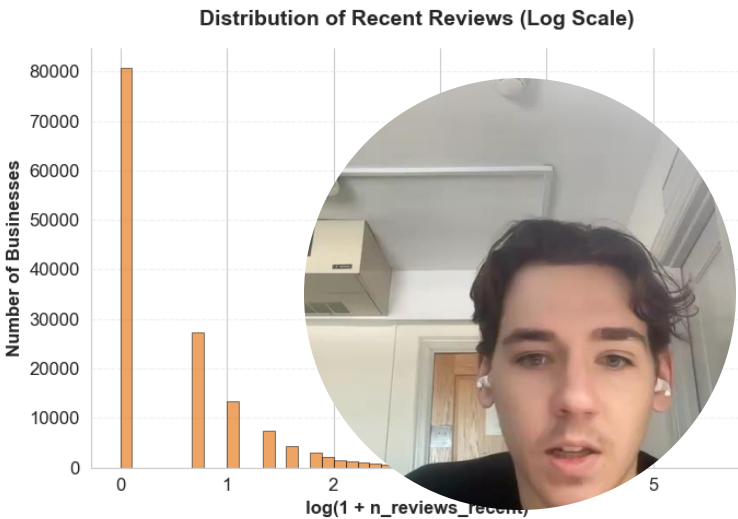
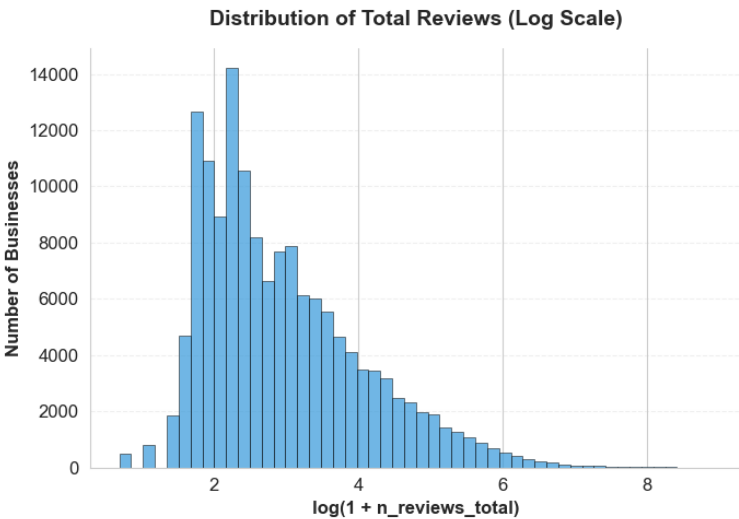
200,100

Pictures

Table 1 displays all 38 variables available on the Yelp dataset. There are 5 .json files in total. Figure 1, shown below Table 1, shows the distribution of all customer reviews as well as recent reviews, defined to be within 180 days before January 1st, 2021.

Table 1: Yelp Open Dataset JSON files and available fields.

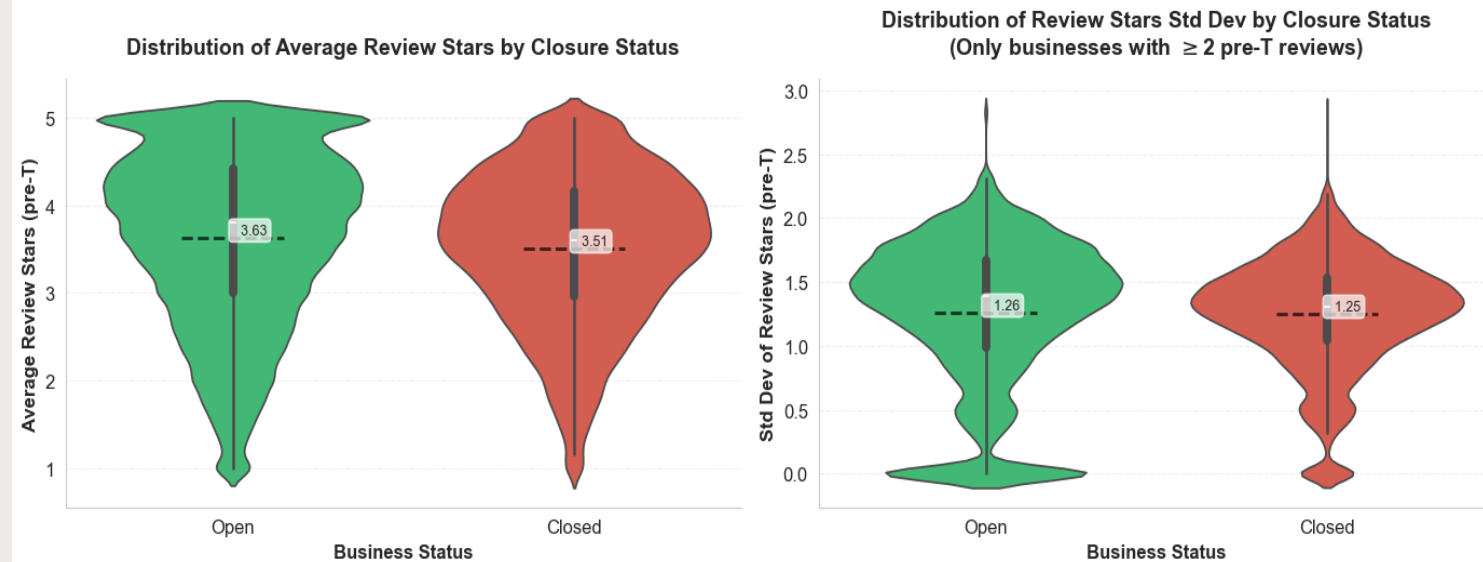
JSON file	Available fields (columns)
yelp_academic_dataset_checkin.json	business_id, date
yelp_academic_dataset_tip.json	business_id, compliment_count, date, text, user_id
yelp_academic_dataset_review.json	business_id, cool, date, funny, review_id, stars, text, useful, user_id
yelp_academic_dataset_business.json	address, attributes, business_id, categories, city, hours, is_open, latitude, longitude, name, postal_code, review_count, stars, state
yelp_academic_dataset_user.json	average_stars, compliment_cool, compliment_cute, compliment_funny, compliment_hot, compliment_list, compliment_more, compliment_note, compliment_photos, compliment_plain, compliment_profile, compliment_writer, cool, elite, fans, friends, funny, name, review_count, useful, user_id, yelping_since



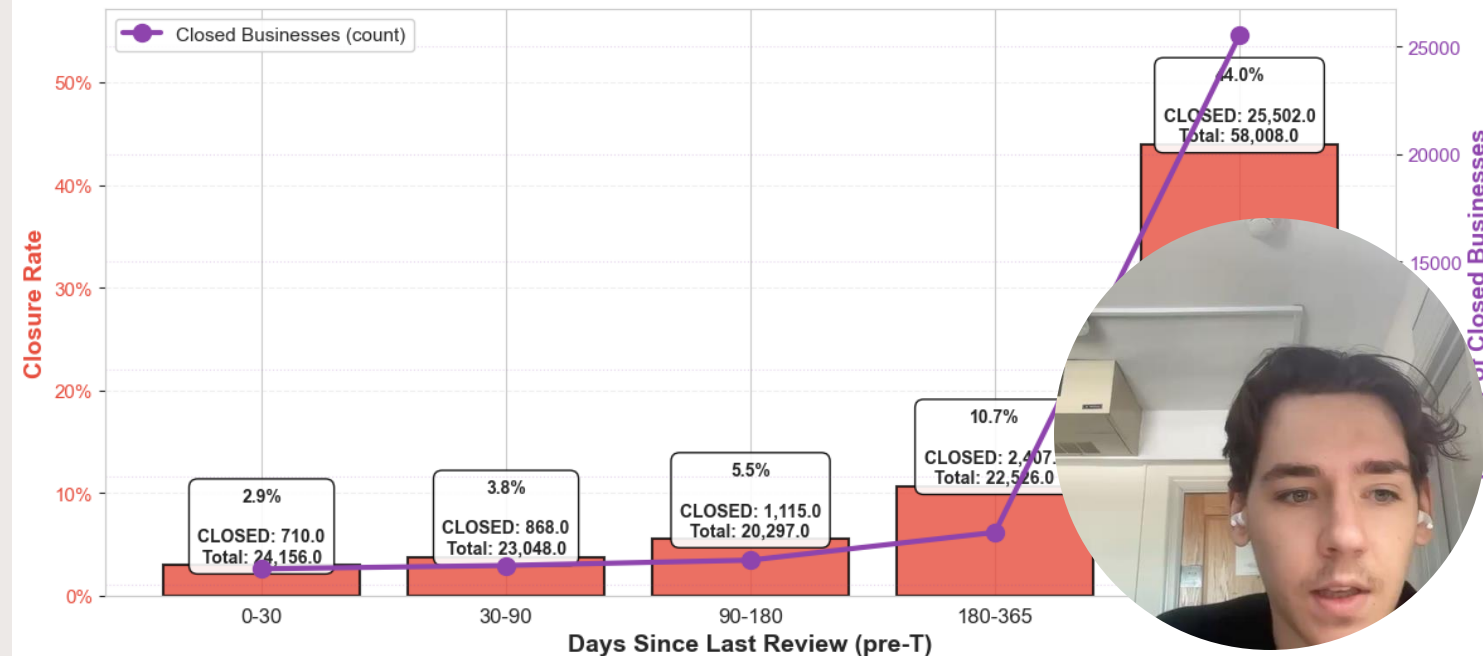
EXPLORATORY DATA ANALYSIS

- If last review > 365 days pre-T, closure rate spiked to about 44%
- Days since last review are highly correlated with business closure rates
- Average star rating distributions appear to be quite similar for Open and Closed businesses
 - Open businesses have a slightly higher average
- Construct all features using only the data before a cutoff T, which I selected to be 12-31-2020
- Classification Target Variable: closed = 1 - is_open
- Regression Target: avg_stars_postT, defined only for businesses that remain open and have ≥ 3 post-T reviews

Figure 2: (Top) Distribution of Average Review Star Ratings, (Bottom) Distribution of Standard Deviation for businesses with ≥ 2 reviews before a defined cutoff, T



Closure Rate and Count by Recency of Last Review (pre-T)



CLASSIFICATION METHODS (CLASSICAL ML)

- Logistic Regression: Good baseline with interpretable coefficients
- Decision Trees: allow for non-linearities and interactions among predictor variables
- Random Forests: similar advantages as those of decision trees, but with an additional reduction to variance

Figure 3(a); Sigmoid Activation Function Illustrated

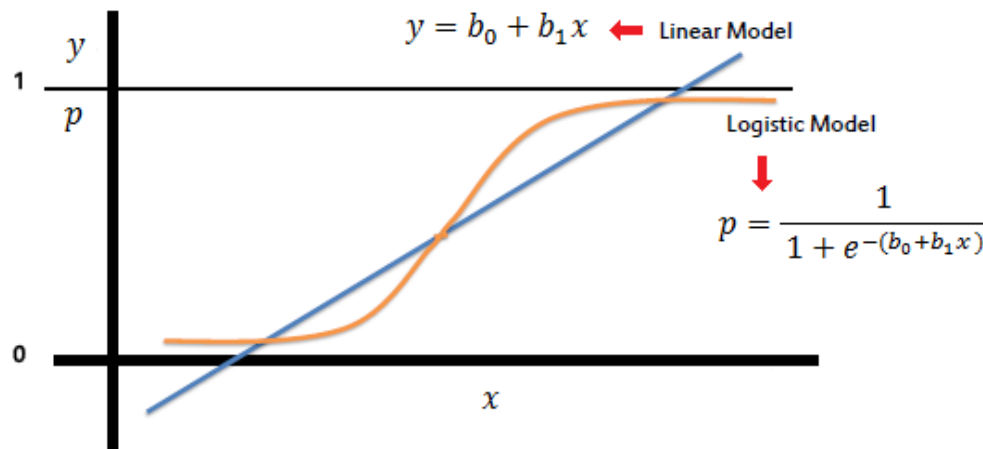


Figure 3(b): (Top)
Additional visual on
the logistic regression
model

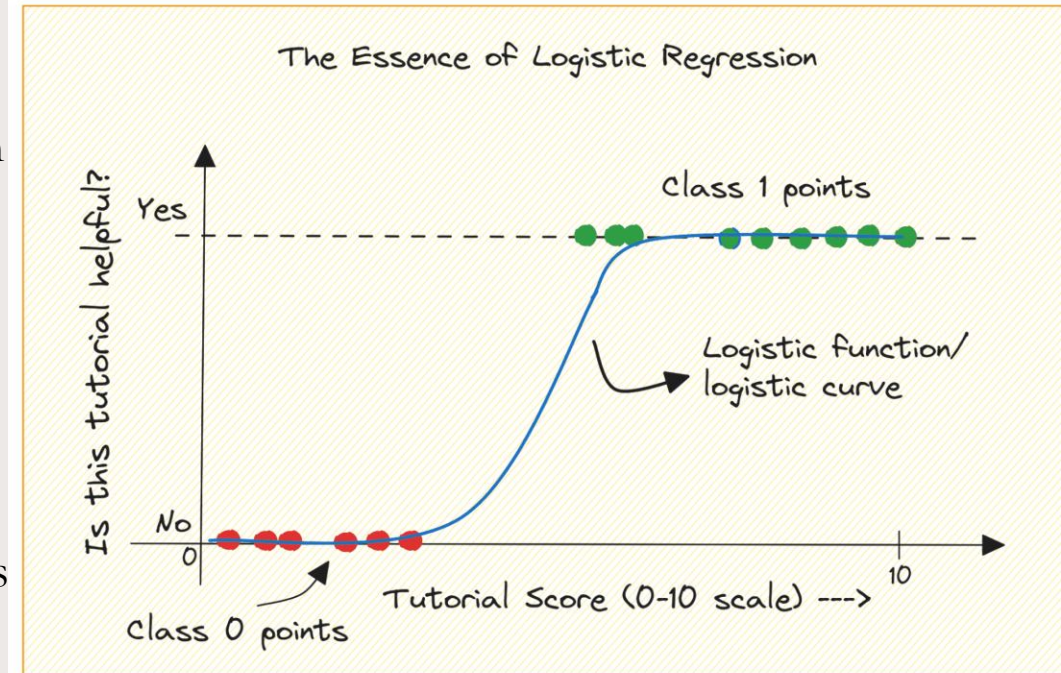
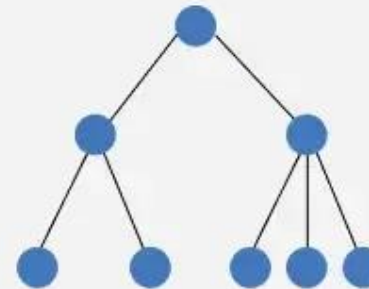


Figure 3(c): (Bottom)
Single Decision Trees
vs. Tree Ensembles

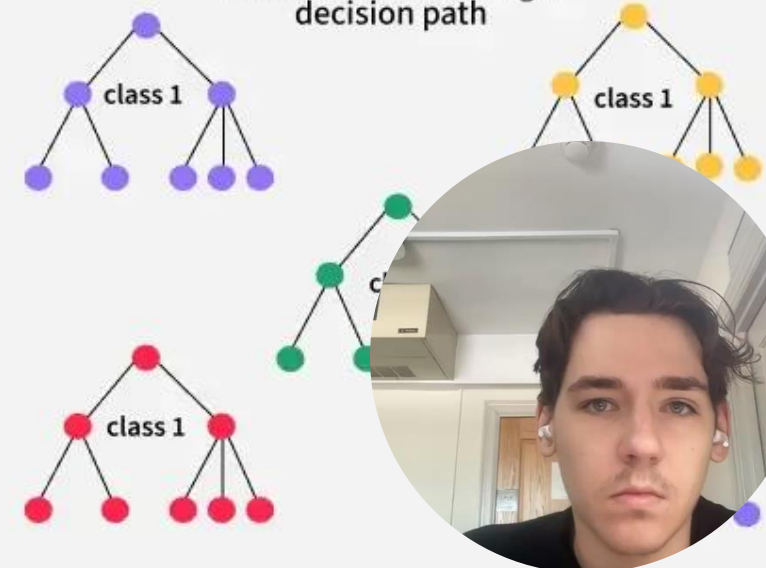
Single Decision Tree

Ensemble of trees for more accurate and robust prediction



Random Forest

Prediction from a single decision path



CLASSIFICATION METHODS (NEURAL NETWORKS)

- 2 Models: MLP-1, MLP-2
- Many inputs, single output (perceptron)
- MLP-1 Pathway: Dense (h1) -> ReLU activation -> Dropout Regularization -> Output
- MLP-2 Pathway: Dense (h1) -> ReLU activation -> Dropout Regularization -> Dense(h2) -> ReLU -> Output
- Classification output: sigmoid, BCE Loss
- Regression output: linear, MSE Loss
- Optimized with Adam, used Early Stopping

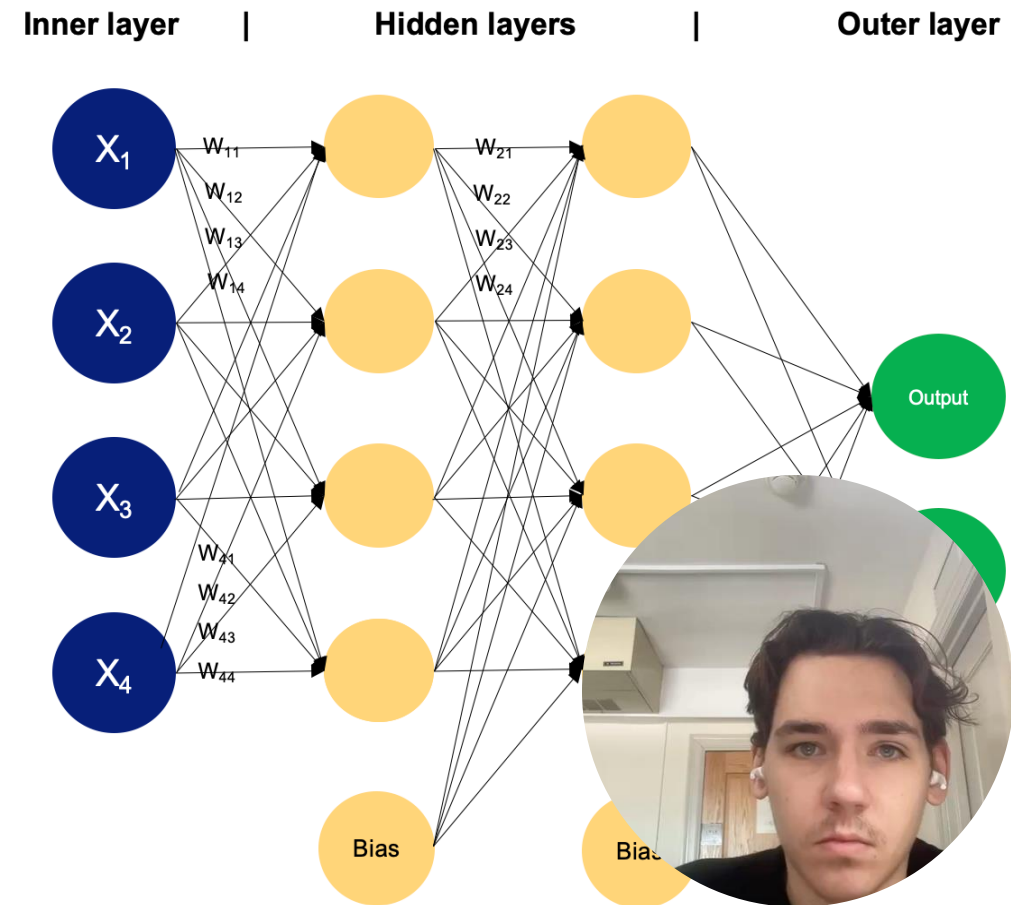
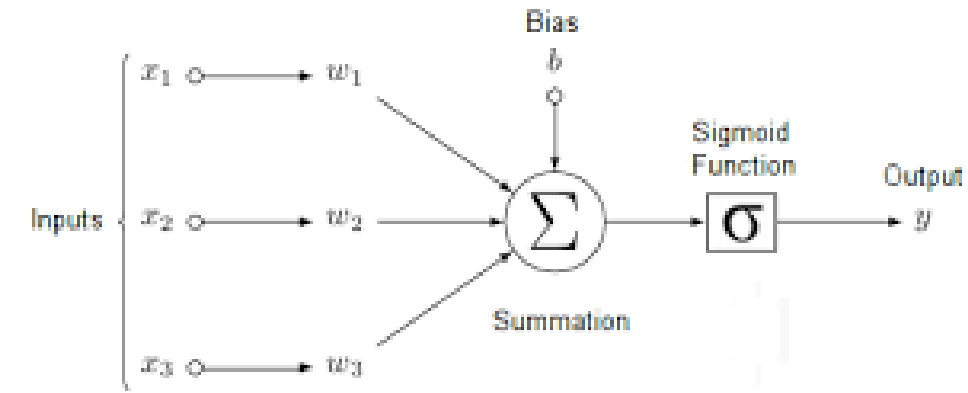
$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Eq'n 4.1: Mean Squared Error Formula
Eq'n 4.2: Binary Cross Entropy Formula

$$BCE = -\frac{1}{n} \sum_{i=1}^n (y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i))$$

Figure 4(a): (top) general structure of a feedforward neural network (NN) with a simple neuron structure

Figure 4(b): (bottom) general structure of a feedforward neural network (NN) with two hidden layers



METRICS (CLASSIFICATION)

Accuracy - Measures overall correctness of the model

- % of correct predictions out of all predictions

Precision - Accuracy of positive predictions

- High precision = less false positives

Recall - Proportion of all positive cases that were identified correctly

- High recall = model finds most relevant instances often

F1 - Combined measure of accuracy using both precision and recall

- How many selected items are relevant AND how many relevant items are selected
- Particularly useful for imbalanced datasets, where accuracy might be misleading on its own
- Good at aiding model selection where we are discriminating between models with different strengths.

ROC-AUC

- Evaluates binary classifier's ability to distinguish between classes
- Plots sensitivity (true positive rate) vs 1-specificity (false positive rate) across different thresholds
- 0.5 – random score, 1 - perfect score

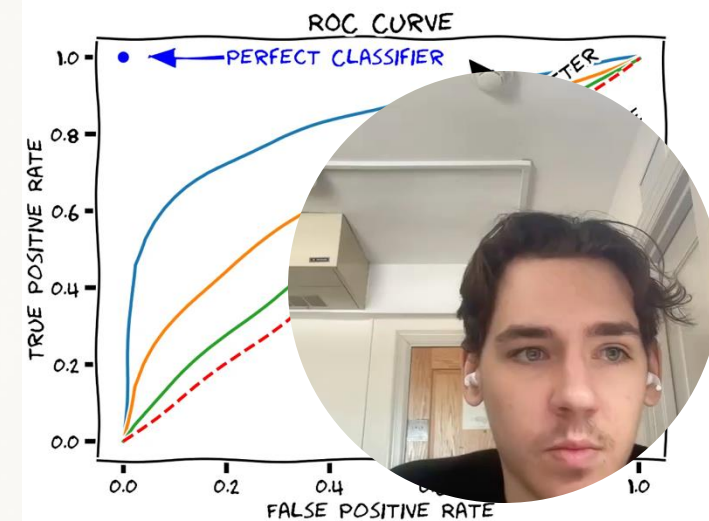
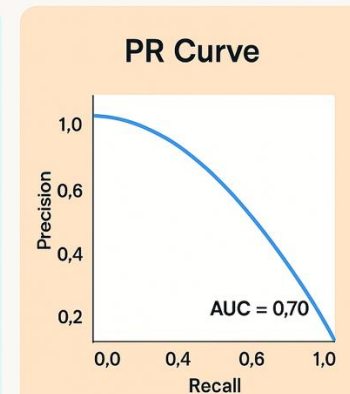
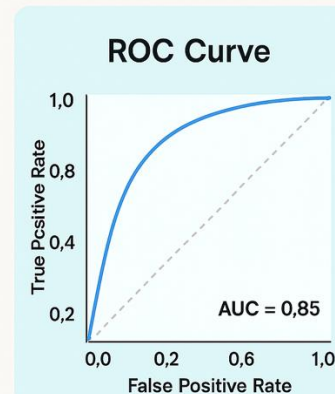
PR-AUC

- Good for imbalanced datasets with are positive class
- Plots precision vs. recall
- Selects models based on identifying positive class effectively

$$\begin{aligned} \text{precision} &= \frac{TP}{TP + FP} \\ \text{recall} &= \frac{TP}{TP + FN} \\ F1 &= \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \\ \text{accuracy} &= \frac{TP + TN}{TP + FN + TN + FP} \\ \text{specificity} &= \frac{TN}{TN + FP} \end{aligned}$$

Figure 5: (a, left) comparison between ROC and PR-AUC curves; (b, right) general shape of an ROC curve

ROC vs PR-AUC



METRICS (REGRESSION)

MSE

- Strong penalty on large errors
- Smooth and differentiable

RMSE

- More interpretable (units same as target)
- Sensible to outliers
- Comparable across models

MAE

- Linear Penalty
- Robust to outliers
- More interpretable than MSE for skewed distributions

R²

- Relative performance compared to a baseline
- Highly interpretable

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

$\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$ are predicted values
 $y_1, y_2, \dots, y_n$ are observed values
 n is the number of observations

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$



HYPERPARAMETER SELECTION (CLASSIFICATION)

- Stratified 5-fold to preserve dataset imbalance
- Search with GridSearchCV
- All hyperparameter values defined to the right. To select the best, choose the best CV score, refit on the training data, and evaluate once on the test data
- RandomizedSearchCV that is 3-fold for Neural Networks

Table 6: Hyperparameter search and cross-validation design for **classification** models.

Task	Model	Search type	CV scheme	Tuned hyperparameters
Classification	Logistic regression (L2)	GridSearchCV	Stratified 5-fold	Regularization strength $C \in \{0.1, 1, 10\}$.
Classification	Decision tree classifier	GridSearchCV	Stratified 5-fold	Tree depth and leaf size: <code>max_depth</code> $\in \{3, 5, 10, \text{None}\}$; <code>min_samples_leaf</code> $\in \{1, 5, 10\}$.
Classification	Random forest classifier	GridSearchCV	Stratified 5-fold	Ensemble size and tree complexity: <code>n_estimators</code> $\in \{100, 300\}$; <code>max_depth</code> $\in \{\text{None}, 10\}$; <code>min_samples_leaf</code> $\in \{1, 5\}$; <code>max_features</code> $\in \{\text{'sqrt'}, \text{'log2'}\}$.
Classification	MLP-1 (NN)	RandomizedSearchCV	3-fold	Network width and regularization: <code>units</code> $\in \{64, 128, 256\}$; <code>dropout</code> $\sim \text{Unif}(0, 0.3)$; $\lambda_2 \sim \text{LogUnif}(10^{-5}, 10^{-2})$; $\text{lr} \sim \text{LogUnif}(10^{-4}, 3 \times 10^{-3})$.
Classification	MLP-2 (NN)	RandomizedSearchCV	3-fold	



HYPERPARAMETER SELECTION (REGRESSION)

- Extremely similar to classification hyperparameter selection
- Narrower hyperparameter space, still perform {5,3}-CV to select the best params

Table 7: Hyperparameter search and cross-validation design for **regression** models.

Task	Model	Search type	CV scheme	Tuned hyperparameters
Regression	Ridge regression	GridSearchCV	5-fold	Regularization strength $\alpha \in \{0.01, 0.1, 1, 10\}$.
Regression	Decision tree regressor	GridSearchCV	5-fold	Tree depth and leaf size: max_depth $\in \{8, 12, None\}$; min_samples_leaf $\in \{1, 5\}$.
Regression	Random forest regressor	GridSearchCV	5-fold	Tree depth and leaf size: max_depth $\in \{8, 12, None\}$; min_samples_leaf $\in \{1, 5\}$.
Regression	MLP-1 (NN, regression)	RandomizedSearchCV	3-fold	Learning rate $\in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$; hidden_layer_sizes $\in \{(10, 10), (10, 20), (20, 10), (20, 20)\}$.
Regression	MLP-2 (NN, regression)	RandomizedSearchCV	3-fold	Learning rate $\in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$; hidden_layer_sizes $\in \{(10, 10), (10, 20), (20, 10), (20, 20)\}$.



RESULTS

(CLASSIFICATION)

- Best overall ROC-AUC - MLP-2
 - Despite being the best in terms of ROC-AUC, increases false positives
- Best accuracy/prevision - Random Forest
- Review Features carry the most predictive power

Table 8: Classification performance on the held-out test set using **ALL** features.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic (L2)	0.755	0.446	0.764	0.563	0.843	0.629
RidgeClassifier (cal.)	0.821	0.664	0.271	0.385	0.777	0.520
Decision Tree	0.756	0.451	0.831	0.585	0.853	0.613
Random Forest	0.862	0.764	0.478	0.588	0.877	0.705
MLP-1 (1 hidden layer)	0.803	0.516	0.786	0.623	0.875	0.700
MLP-2 (2 hidden layers)	0.816	0.538	0.778	0.636	0.879	0.716

Table 9: Ablation over feature groups on the test set using the **best classifier per group** (model shown).

Feature group	Best model	ROC-AUC	PR-AUC
GEO	Random Forest	0.607	0.301
BUSINESS	Logistic (L2)	0.704	0.352
REVIEW	Random Forest	0.862	0.651
ALL	Random Forest	0.877	0.705

Note: Best models were determined by highest ROC-AUC values, and the ties were broken by Pre



REGRESSION RESULTS

- Predicting average star ratings are harder due to the discretization of ratings, missing review-derived sentiment features, etc.
- Only done on businesses with more than 3 reviews dated after T to remove noisy businesses with none or zero reviews from the dataset.
- Higher R^2 was on the order of 53%, so could only explain 53% of the variance
- RF best overall, MLP-1 came in close second

Table 12: Top positive and negative coefficients from the L1-logistic regression (standardized features). Positive values increase the log-odds of closure; negative values decrease it.

Top increases (positive β)		Top decreases (negative β)	
Feature	Coefficient	Feature	Coefficient
cat_Vegetarian	2.554	num_n_reviews_recent	-2.645
cat_Asian Fusion	2.391	cat_Religious Organizations	-2.190
cat_Vietnamese	2.346	cat_state_FL	-1.954
cat_Middle Eastern	2.332		
cat_Korean	2.311		
cat_Barbeque	2.296		
cat_Mediterranean	2.204		
cat_Cuban	2.201		
cat_Greek	2.189		
cat_Indian	2.174		

Table 11: Regression performance on the held-out test set (ALL features).

Model	MSE	RMSE	MAE	R^2
Ridge Regression (best α)	0.6647	0.8153	0.6275	0.5200
Random Forest (best params)	0.6518	0.8074	0.6210	0.5293
MLP-1	0.6565	0.8103	0.6205	0.5259
MLP-2	0.6988	0.8359	0.6528	0.4954

Notes: Target = post- T average star rating for businesses with ≥ 3 post- T reviews. Preprocessing is identical for all models. No explicit feature selection is used for regression.



DISCUSSION

- Data set includes 2020: COVID-19 could have been a large confounding variable that swayed business closures
- The spatial time design is good for controlling data leakage, but limits overall signal



- The problem of predicting business closure overall is much more predictable, with RF achieving the highest performance
- The average star rating task was only moderately predictable, with average Coefficient of Determination, RMSE, and related metrics showing equally average performance.
- Possible next steps: text embeddings account for COVID-19 confounding

